Project Design Phase-I
Solution Architecture

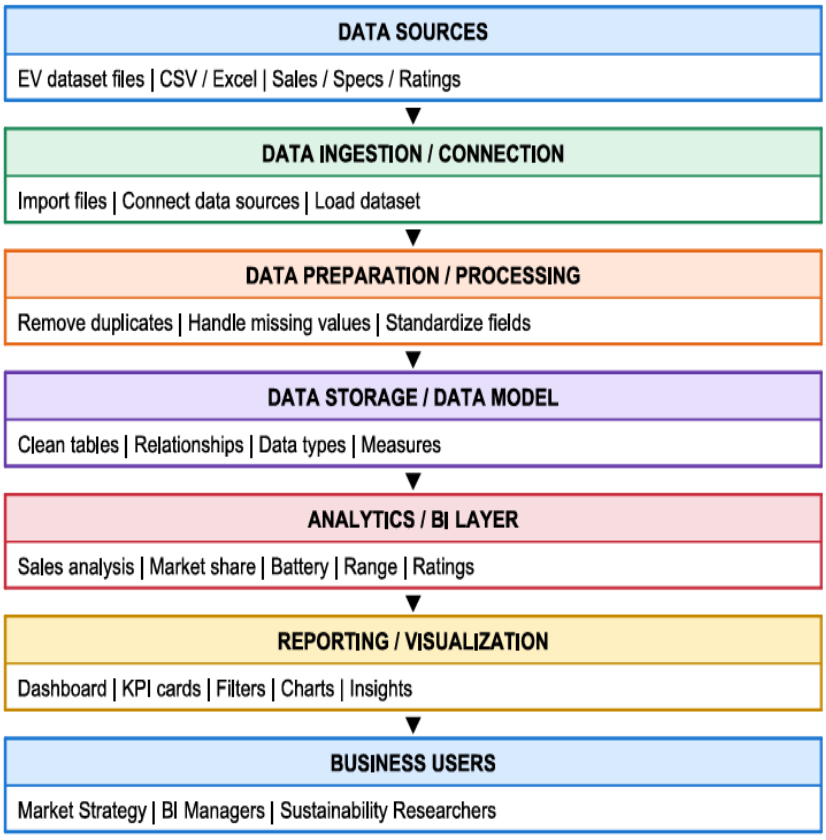
Date	21/10/2026
Team ID	Suhani Porwal
Project Name	**Electric Vehicle Market Performance & Insights**
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture for the **Electric Vehicle Market Performance & Insights** project describes how EV market data is collected, prepared, stored, analyzed, and presented through an interactive dashboard. It connects raw data to useful insights for market strategy teams, automotive business intelligence managers, and sustainable mobility researchers.

- Bring together EV sales, manufacturer, battery, range, manufacturing-country, and customer-rating data.
- Clean and transform the data to handle missing values, inconsistent categories, and data-quality issues.
- Create analysis measures such as annual sales, market share, battery technology distribution, average driving range, and customer ratings.
- Present results through clear charts, KPIs, filters, and comparisons so users can make data-informed decisions.

Sample Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technology Used
Data Sources	Stores EV market records such as sales, vehicle category, manufacturer, battery specifications, range, country, and ratings.	CSV / Excel / project dataset
Data Preparation	Cleans missing or duplicate records, standardizes fields, and prepares data for analysis.	Python (Pandas) or Power Query
Data Model & Analytics	Defines relationships and calculates KPIs such as annual sales, market share, range, and ratings.	Power BI measures / Excel / Python
Dashboard & Visualization	Displays interactive charts, KPI cards, filters, and comparisons for the three user scenarios.	Power BI / Tableau / Excel dashboard